

MEHREEN ASGHAR

AI Engineer • Backend & LLM Application Developer • Freelance Software Engineer

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PROFILE

Software Engineering student and freelance-ready AI/backend developer with hands-on experience building production-style REST APIs and integrating Large Language Models into real applications. Recently designed, built, and deployed a complete FastAPI backend from scratch — covering authentication, database design, automated testing, CI/CD, and structured LLM-powered features — in addition to an AI internship with the AIRI team at Arfa Software Technology Park (PITB). Comfortable working independently across the full stack, from database schema to client-facing API, and available to take on freelance projects in API development, backend engineering, and applied LLM integration.

FREELANCE SERVICES

- Backend API development with FastAPI / Python (CRUD systems, authentication, database design)
- JWT-based authentication & authorization systems for web and mobile apps
- Integrating LLMs (Gemini, OpenAI, Claude) into applications with structured, validated outputs
- Database design & migrations with PostgreSQL, SQLAlchemy, and Alembic (zero-downtime schema changes)
- Automated testing (Pytest) and CI/CD pipeline setup (GitHub Actions) for reliable, production-ready delivery
- Bug fixing, code review, and refactoring for existing Python/JavaScript codebases

FEATURED PROJECT

Production-Style REST API with JWT Auth, Testing, CI/CD & LLM Integration

Personal Project

Python • FastAPI • PostgreSQL • SQLAlchemy • Alembic • Pytest • GitHub Actions • Google Gemini API

- Designed and built a full-featured blog-style REST API from the ground up: user accounts, posts with full CRUD, a many-to-many voting/likes system, and pagination with search.
- Implemented secure JWT authentication (password hashing, protected routes, per-user post ownership enforcement) so only a post's owner can edit or delete it.
- Managed all schema changes through Alembic migrations, enabling safe, zero-downtime updates to the database without ever losing existing data.
- Wrote a Pytest suite (users, auth, and post-ownership tests) using isolated test fixtures and a dedicated test database, then wired it into a GitHub Actions CI/CD pipeline that automatically runs on every push.
- Integrated Google's Gemini API to add an AI-powered post-analysis feature: content is sent to the LLM and a strict JSON schema (via Pydantic) forces the model to return a validated summary, sentiment, and tag list, which is then persisted back to the database.
- Added resilient error handling for external API calls, including automatic retry logic for transient LLM service outages.
- Repository: github.com/mehreen18/Api_development

EXPERIENCE

AI Intern — AIRI Team

Arfa Software Technology Park, under Punjab Information Technology Board (PITB)

- Worked with the AIRI team on applied AI/ML tasks, gaining hands-on exposure to real-world project workflows in a professional tech environment.
- Collaborated with engineers on AI-related development work within a government-backed technology ecosystem, strengthening practical skills in Python, ML concepts, and team-based problem solving.

OTHER PROJECTS

Student Management System

- Developed a full CRUD-based application to manage student records using core programming and OOP principles, with a focus on clean, modular code.

React Web Application

- Built a responsive web application using React.js with reusable components and clean state management.

AI & Machine Learning Practice Projects

- Applied Python to hands-on AI/ML exercises, exploring generative AI, LLM applications, and the basics of AI agent design.

TECHNICAL SKILLS

Backend / APIs	Python, FastAPI, REST API design, JWT authentication, Pydantic, SQLAlchemy, Alembic
Databases	PostgreSQL, SQL (schema design, migrations, relationships)
Testing & DevOps	Pytest, GitHub Actions (CI/CD), Git & GitHub, environment configuration
AI / LLM	LLM fundamentals, prompt engineering, structured outputs, Google Gemini API, RAG concepts, LangChain basics, Hugging Face ecosystem, OpenAI / Anthropic APIs
Machine Learning	ML fundamentals, NumPy, Pandas, Scikit-learn; introductory TensorFlow & PyTorch
Frontend	JavaScript, React.js, HTML, CSS
Other Languages	C++, Java
Currently Building Toward	Docker, vector databases, cloud basics (AWS/Azure), MLOps fundamentals, full-stack AI app deployment

EDUCATION

Bachelor of Software EngineeringIn Progress

Superior University (Gold Campus)

CERTIFICATIONS

- Full Stack Development Training

SOFT SKILLS

Problem Solving • Communication • Teamwork • Quick Learner • Time Management • Analytical Thinking • Independent Client Delivery

LANGUAGES & INTERESTS

Languages: English, Urdu

Interests: Artificial Intelligence, Machine Learning, Generative AI, AI Agents, Software Development